

LONDON WC PENDEREL HOUSE, United Kingdom

WMO: 037780

Lat: 51.517N

Lon: 0.117W

Elev: 75

StdP: 14.66

Time Zone: 0.00 (EUW)

Period: 82-92

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	27.3	30.2	20.0	15.1	31.1	22.4	16.9	33.4	31.4	49.2	29.1	43.6	13.0	40	0.605

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.0	80.9	66.0	78.1	64.4	75.3	63.4	67.8	78.5	66.0	75.0	64.5	72.7	9.7	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.6	88.1	72.2	62.3	84.1	70.8	61.0	80.3	69.1	32.1	78.6	30.7	75.1	29.6	72.4	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
26.2	23.1	21.0	DB	26.2	86.1	5.6	4.6	22.1	89.3	18.9	92.0	15.7	94.6	11.7	97.9
			WB	23.9	71.1	5.1	2.7	20.3	73.0	17.3	74.6	14.4	76.1	10.7	78.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.0	42.5	41.8	46.4	49.4	55.4	60.4	65.9	65.2	60.5	55.0	47.6	45.4
	DBStd	9.77	6.12	6.72	4.97	5.32	5.35	5.24	4.83	4.82	4.48	4.40	5.40	5.06
	HDD50	963	235	236	132	74	10	0	0	0	0	8	109	159
	HDD65	4547	697	650	576	469	302	159	48	59	146	311	521	609
	CDD50	2074	3	6	21	56	178	313	492	472	316	163	38	16
	CDD65	181	0	0	0	0	4	23	75	66	12	1	0	0
	CDH74	891	0	0	0	0	24	106	431	290	38	2	0	0
	CDH80	141	0	0	0	0	1	10	82	47	1	0	0	0
(14) Wind	WSAvg	11.2	12.8	12.8	12.2	11.6	10.7	10.4	10.1	9.7	9.8	10.9	11.1	11.9
(15) Precipitation	PrecAvg	23.1	2.0	1.3	1.8	1.7	1.9	1.9	1.7	1.9	2.2	2.4	2.2	2.1
	PrecMax	31.5	5.0	3.5	3.9	4.2	3.5	4.2	3.6	4.6	5.7	5.4	5.5	4.4
	PrecMin	17.2	0.4	0.3	0.2	0.4	0.0	0.2	0.4	0.3	0.2	0.1	0.7	0.5
	PrecStd	3.4	1.1	0.7	0.9	0.9	1.0	1.2	0.9	1.1	1.5	1.7	1.2	1.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.5	57.7	64.0	70.0	77.1	81.5	87.0	85.2	78.2	70.5	60.9	57.5
		MCWB	51.6	51.4	53.7	56.3	62.5	67.5	68.2	68.3	63.9	61.3	55.9	53.3
	2%	DB	53.6	54.6	58.4	65.6	72.3	76.9	81.9	79.9	74.2	65.9	59.1	56.1
		MCWB	50.3	49.9	50.8	54.1	59.9	63.3	66.3	65.4	63.0	59.6	55.3	52.5
	5%	DB	52.4	52.7	55.8	61.7	68.6	73.6	78.8	77.0	71.0	63.6	57.4	54.8
		MCWB	49.1	48.4	49.4	52.3	57.6	62.3	64.8	64.0	61.1	57.7	54.1	51.5
	10%	DB	50.9	50.9	53.4	58.4	65.0	70.0	75.6	74.3	68.0	61.6	55.5	53.1
		MCWB	47.7	47.0	47.9	50.3	55.5	60.4	63.5	63.2	59.8	56.4	52.3	50.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.8	52.9	54.2	58.4	63.8	68.7	70.1	69.6	66.5	62.3	57.5	54.6
		MCDB	54.8	56.2	61.6	66.7	74.4	78.6	83.2	83.2	74.6	67.9	60.1	56.5
	2%	WB	51.1	50.9	52.2	55.2	60.8	65.9	68.1	67.3	63.9	60.5	56.1	53.2
		MCDB	53.2	54.0	57.4	63.8	71.2	74.3	79.6	77.2	70.6	65.3	58.6	55.4
	5%	WB	49.7	49.1	50.6	53.2	58.5	63.4	66.2	65.5	62.5	58.9	54.4	51.9
		MCDB	52.0	52.2	54.6	60.2	67.6	71.5	75.9	74.0	68.8	62.3	57.0	54.5
	10%	WB	48.0	47.4	49.1	51.2	56.5	61.2	64.4	64.2	61.2	57.5	52.7	50.3
		MCDB	50.8	50.5	52.7	57.7	63.9	68.9	73.2	71.9	67.1	60.9	55.4	53.0
(35) Mean Daily Temperature Range	5% DB	MDBR	7.3	7.7	9.8	12.2	12.8	12.7	14.0	13.4	11.6	9.5	7.8	6.8
		MCDBR	8.4	9.4	12.6	16.9	18.6	17.9	19.4	17.6	16.2	11.0	8.2	7.4
		MCWBR	8.2	7.2	7.6	8.8	9.4	9.1	8.6	7.7	7.6	6.4	6.4	6.9
	5% WB	MCDBR	8.7	9.2	11.1	15.5	17.6	16.8	18.0	15.9	13.8	10.4	8.1	7.6
		MCWBR	9.0	8.1	7.6	8.8	9.8	9.1	8.9	7.6	8.0	7.6	7.0	7.4
(40) Clear-Sky Solar Irradiance	taub		0.331	0.347	0.384	0.405	0.410	0.411	0.407	0.397	0.379	0.362	0.335	0.321
	taud		2.369	2.373	2.264	2.225	2.251	2.276	2.294	2.340	2.389	2.419	2.414	2.355
	Ebn at Noon		213	244	254	263	268	268	267	264	255	235	211	196
	Edh at Noon		20	26	35	40	41	40	39	36	30	24	19	17
(44) All-Sky Solar Radiation	RadAvg		254	463	810	1269	1527	1691	1603	1334	1004	580	304	199
	RadStd		22	64	108	138	102	157	153	107	84	53	35	25

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (81 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page